SEAL: Link Prediction Based on Graph Neural Networks

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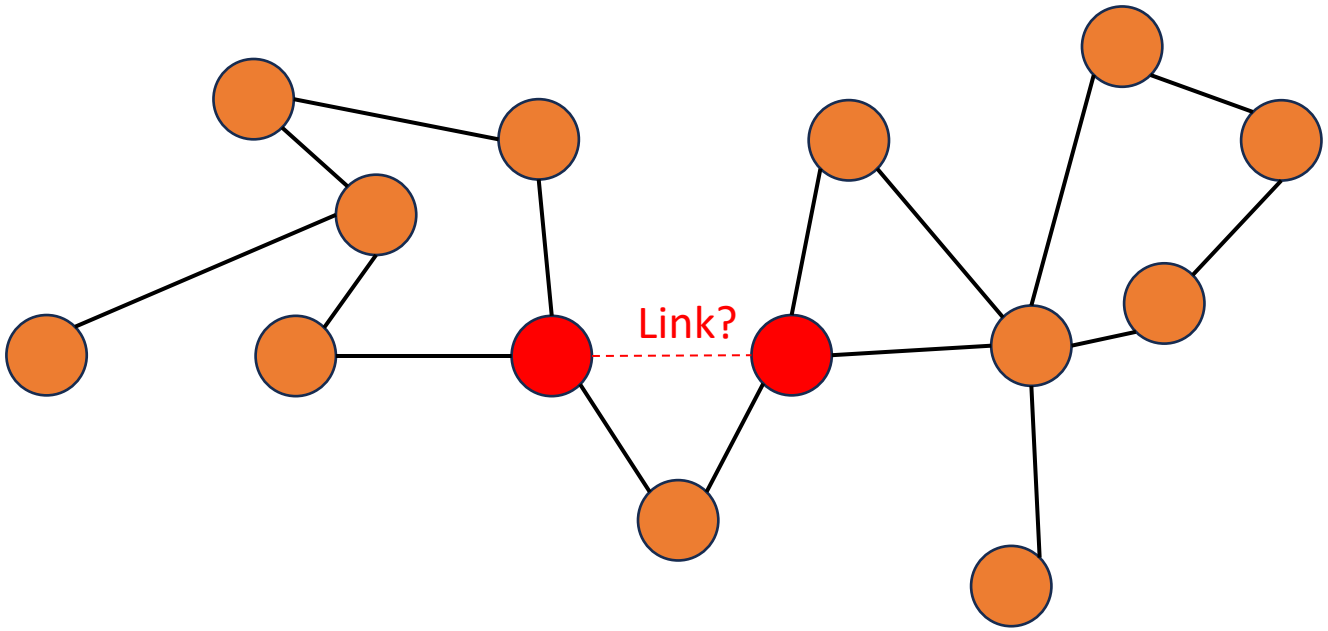
Link Prediction Problem



Applications in Recommendation systems, Knowledge graph completion, metabolic network reconstruction, etc.

Heuristics Based Approach

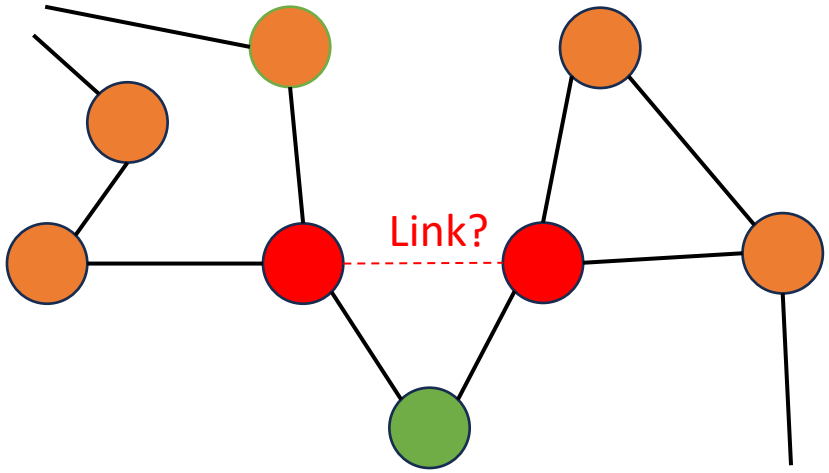
Compute a Node similarity score based on some heuristic related to the neighboring nodes.

Heuristic Degree is classified based on the size of the neighborhood (in hops).

First order (1-hop) heuristics:
Common Neighbor, Preferential attachment

Second Order (2-hop) heuristics:
Adamic-Adar (AA), Resource Allocation

High Order (k-hop k>3) heuristics:
Katz, Rooted PageRank, SimRank



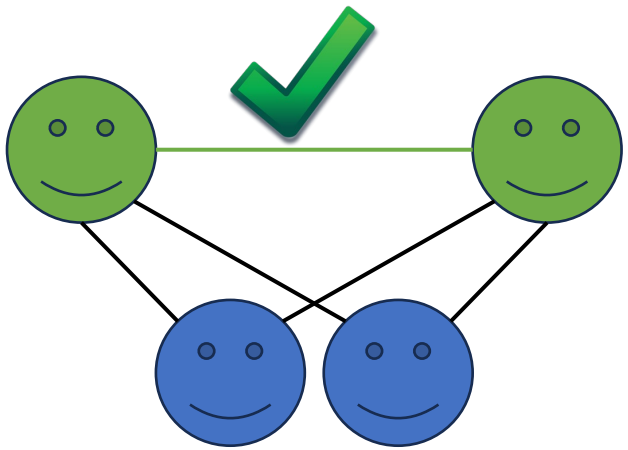
Common Neighbor Heuristic

Limitations with Heuristic Based approach

Requires a lot of Trial & Error to find the right heuristic for a particular network.

E.g., Common Neighbor heuristic works well in social networks but fails in protein-protein interaction (PPI) networks

This makes the approach non-generalizable since every network will require its own heuristic.



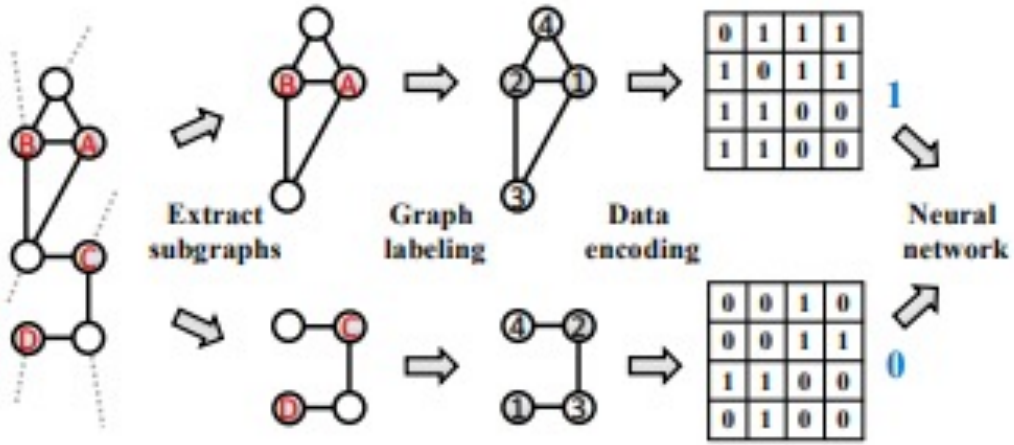
Using DNN to automatically study graph structure heuristics

- First architecture to use this method – Weisfeiler-Lehman Neural Machine

- Achieved state of the art prediction performance

- Uses Weisfeiler-Lehman graph labeling to encode structural information in node label order

- Limitations:
- Uses DNN instead of GNN which misses out on attribute features of nodes & limits learning of graph features.



First Contribution: γ −decaying theory

$Assertion$: $High\ order\ \gamma$ −decaying heuristics can be approximated using a smaller h-hop neighborhood

Preliminaries:

$Theorem$: $h - order\ heuristic\ for\ (x, y)\ can\ be\ calculated\ using\ an\ enclosing\ h - hop\ subgraph\ denoted\ by\ G_{x,y}^{h}\ which\ contains\ the\ h - hop\ neighbourhoods\ of\ x\ \&\ y$

Def: $A\ \gamma - decaying\ heuristic\ for\ (x, y)\ has\ the\ form$:

$$H(x, y) = \eta \sum_{l=1}^{\infty} \gamma^{l} f(x, y, l)\ ;\ \gamma \epsilon [0,1], f\ is\ a\ nonnegative\ function\ of\ x, y, \&\ l$$

$Theorem$: $A\ \gamma - decaying\ heuristic\ H(x,y) = \eta \sum_{l=1}^{\infty} \gamma^l f(x,y,l)\ \ can\ be\ approximated\ using\ G_{x,y}^h$ $and\ the\ approximation\ error\ decreases\ by\ \theta\left(e^h\right)\ if\ f\ statisfies\ the\ following\ properties$:

1. $f(x,y,l) \leq \lambda^l\ where\ \lambda < \frac{1}{\gamma}; and$
2. $f(x,y,l)\ is\ calculable\ from\ G_{x,y}^h\ \forall l \in \{1,2,\dots,g(h)\}, where\ g(h) = ah + b, \&\ a,b \in \mathbb{N}, a > 0$

$Proof$: $Let\ the\ approximated\ heuristic\ \mathcal{H}(x,y) = \eta \sum_{l=1}^{g(h)} \gamma^l f(x,y,l)\,,$

$then\ the\ approximation\ error\ can\ be\ shown\ as$

$$|H(x,y) - \mathcal{H}(x,y)| = \eta \sum_{l=g(h)+1}^{\infty} \gamma^l f(x,y,l) \leq \eta \sum_{l=ah+b+1}^{\infty} \gamma^l \lambda^l = \eta(\gamma\lambda)^{ah+b+1}(1-\gamma\lambda)^{-1}$$

$Therefore, a\ large\ a\ value\ and\ a\ small\ \gamma\lambda\ value\ will\ lead\ to\ faster\ error\ reduction$

Proving that Katz index is $\gamma - decaying$

$Katz\ Index\ is\ defined\ as\ Katz_{x,y} = \sum_{l=1}^{\infty} \beta^l |walks^{<l>}(x,y)| = \sum_{l=1}^{\infty} \beta^l |[A^l](x,y)|\ where, walks^{<l>}(x,y)$

$is\ the\ set\ of\ length\ l\ walks\ between\ x\ \&\ y, and\ A^l\ is\ the\ adjacency\ matrix\ to\ the\ lth\ power$

$Lemma: Any\ walk\ between\ x\ \&\ y\ with\ length\ l \leq 2h+1\ is\ included\ in\ G_{x,y}^h$

$Proof: Let\ w = < x, v_1, v_2, \dots, v_{l-1}, y >\ be\ a\ walk\ of\ length\ 2h+1, so, for\ any\ v_i \in w, either\ d(v_i, x)$
$\leq h\ ord(v_i, y) \leq h, and\ therefore, v_i \in \{h - hop\ neighborhood\ of\ x\ or\ y\}\ which\ means\ v_i \in G_{x,y}^h$

$Setting\ \gamma = \beta\ \&\ \eta = 1\ and\ f(x,y,l) = \left|walks^{<l>}(x,y)\right|, by\ lemma,$
$\left|walks^{<l>}(x,y)\right|\ is\ calculable\ from\ G_{x,y}^h\ and\ so\ it\ satisfies\ property\ 2.$

$It\ satisfies\ property\ 1\ because$
$\left|walks^{<l>}(x,y)\right|\ is\ bounded\ by\ d^l\ where\ d\ is\ the\ maximum\ node\ degree\ of\ the\ network;$

$Setting\ \lambda = d\ yields\ a\ \gamma - decaying\ heuristic\ as\ long\ as\ d < \frac{1}{\beta}, in\ most\ cases\ \beta$
$\leq 5e - 4\ and\ the\ condition\ is\ fulfilled.$

Important Conclusion

The paper contains proofs regarding PageRank & SimRank being $\gamma - decaying\ heuristics$

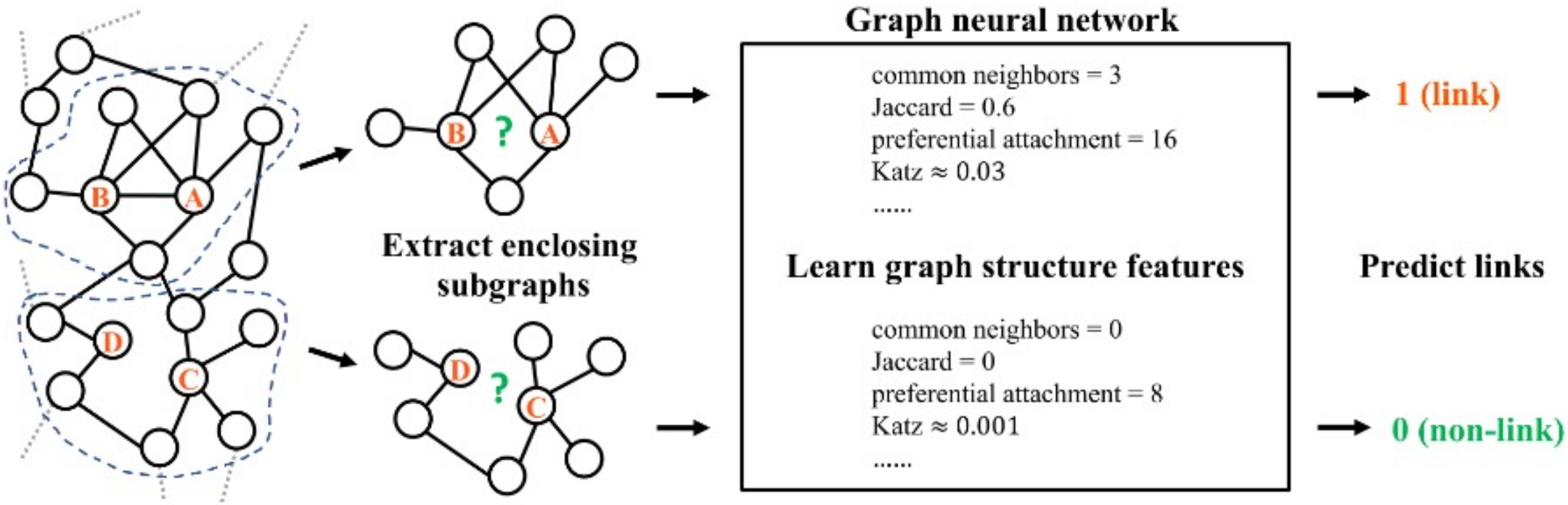
Therefore, high hop neighborhoods are not required for approximating high order heuristics and the neural networks can utilize low h-hop neighborhoods to yield high performance.

SEAL: Implementation

3 step process:

1. Subgraph Extraction
2. Node Information Matrix Construction
3. GNN Training

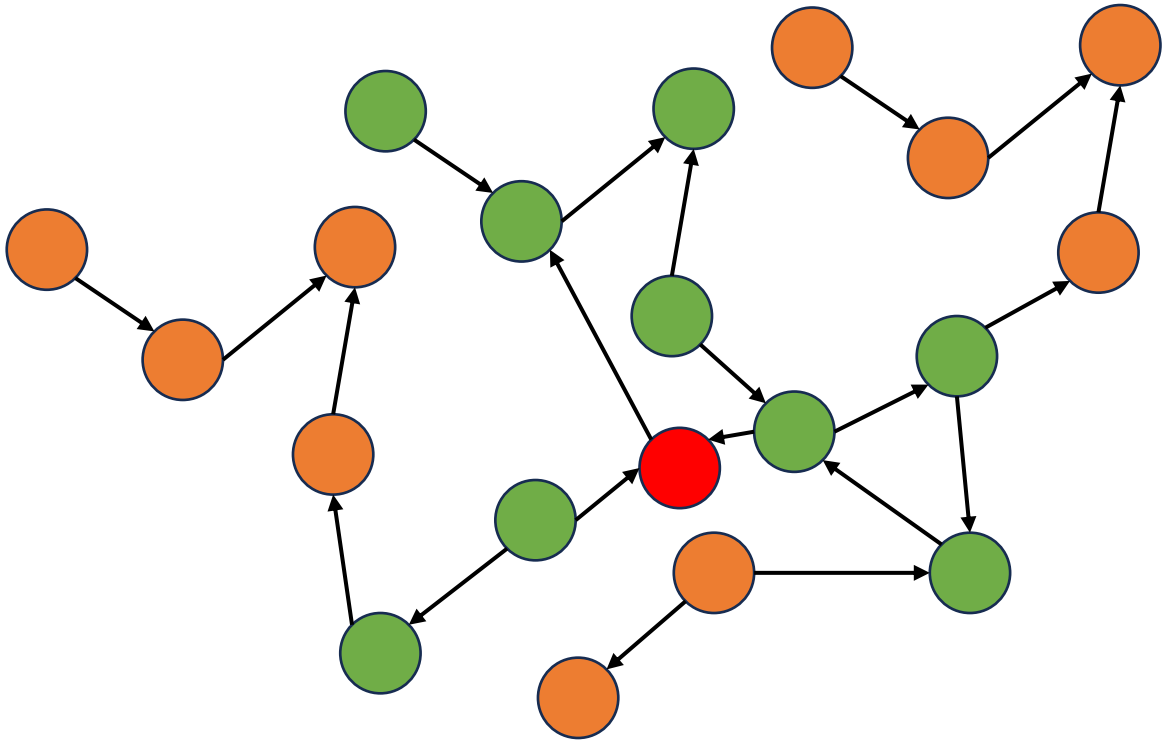
SEAL Execution Overview



Subgraph Extraction

Capture the h-hop neighborhood of Both the link prediction nodes and union the sets.

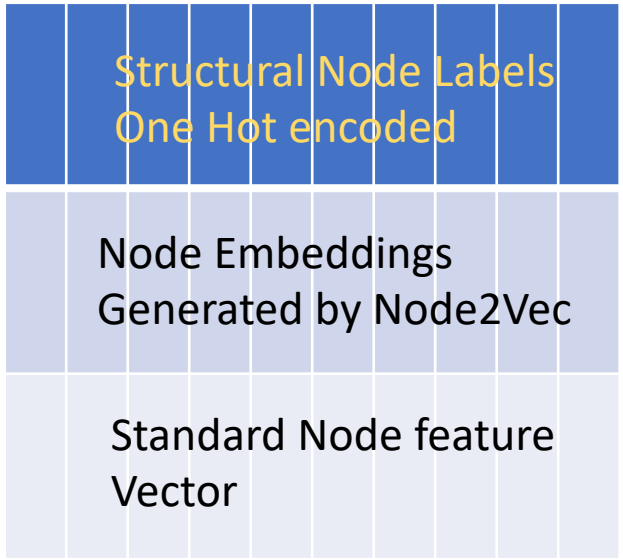
Can be extracted using BFT.



Node Information Matrix Construction

3 Main Components:

1. Structural Node Labels
2. Node Embeddings
3. Node Attributes

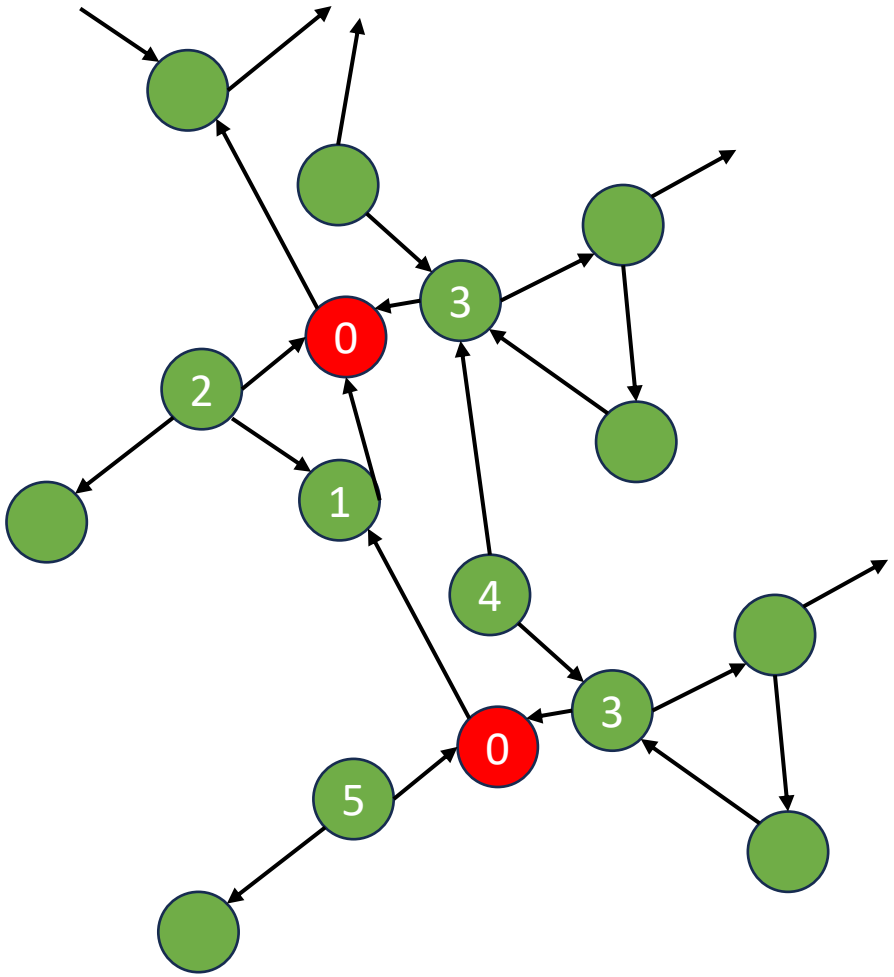


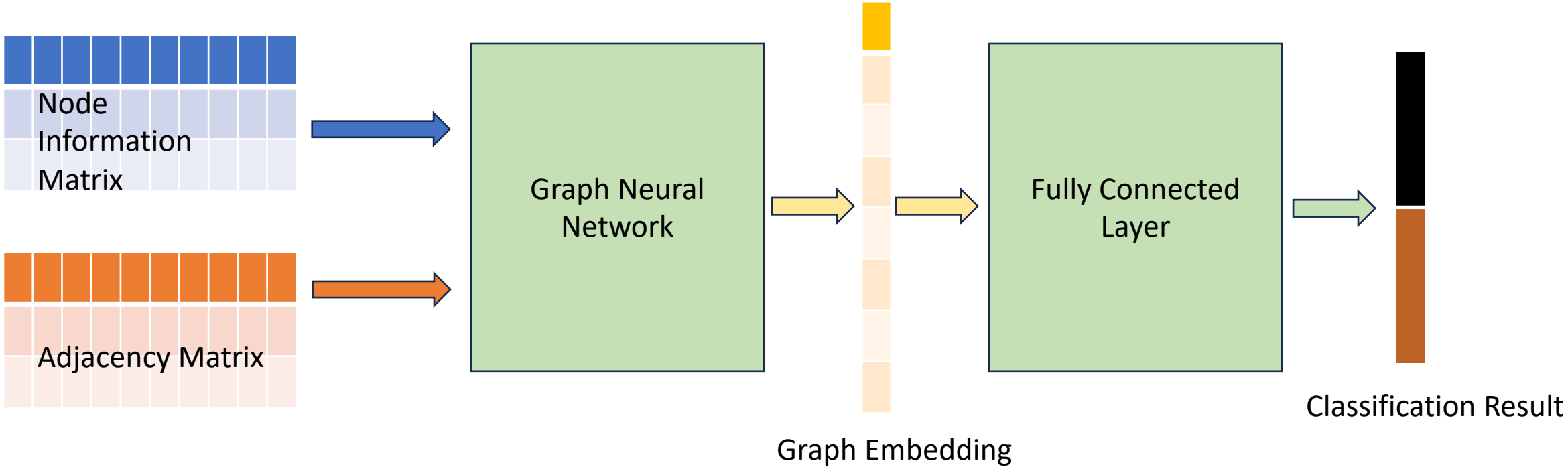
Structural Node Labeling

Encodes information regarding the topology of the subgraph.
Assign labels based on distance from both the central nodes.

Novel Methodology and **second Contribution:**
Double Radius Node Labeling (DRNL)

DRNL uses a function f: $\mathbb{R}^2 \rightarrow \mathbb{R}$ which maps from (d(i,x), d(i,y)) to labeling 1,2,…,k $\forall i \in subgraph$; where d(m,n) is the distance between the nodes m & n



GNN

SEAL uses a standard GNN and doesn't make any innovations in this domain.
The Node Information Matrix and the Adjacency Matrix are passed as inputs to the GNN, which processes them through some convolution layers, followed by aggregation layers which generate a graph embedding vector, which is sent to a fully connected classifier.